

Design and Implementation of Fuzzy-PI gyro-damping Controller for Ballistic

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Abstract—The aim of this research is to propose a hybrid Fuzzy-PI Gyro-Damping (HFPGD) controller applying for autopilot homing missile. The system error (ER) and its change of error (CER) are the inputs of the controller to build the rule-base and the output would be the combination of initial and scaling factor coefficients. This controller was applied to a case study of short-range ballistic missile with maximum speed approximate Mach 3. The results indicate the considerable improvement in stability and overshoot performance between HFPGD controller vice versa traditional PID and Three-loop controller.

Keywords—Fuzzy-PID gyro-damping controller, homing missile, PID controller, three-loop autopilot

I. INTRODUCTION

Guided weapons have been developed for a long time and became more intelligent. There are many types of missile as inertial, programmed or direct guided missile. However homing missile is one of the most common usage type because of its flexible characteristics in chasing and catching the target. This type of missile is guided to target by seeker and onboard computer. Its advantages are the ease to check target positions, and pre-determined direction without depending on an external guidance system. The homing missile has three types: active, semi-active and passive homing missile. In reality, there are various nonlinear and uncertain factors affecting to the performance of homing missiles such as non-linear aerodynamics and environment. Thus, to solve this problem, the controller integrating fuzzy logic is proposed and demonstrated potential results.

The basic guidance method based on proportional navigation (PN) law applied to homing missile was presented as in [1]. The references [2], [3] contributed dynamics model of flying object for modeling derivative motion equations by applying Newton's law and kinematic equations in different kinds of coordinates system. Bulent Ozkan [4] showed generally the modeling method for homing missile and compared the effect of acceleration and angle guidance on short-range air-to-surface missile, a type of homing missile [5]. A popular autopilot using PI/PID or three-loop structure in missile direction were introduced and implemented as in [6] and [7]. Basically, the fuzzy logic was developed and applied for uncertain systems and non-linear model. The overview of control method combining Fuzzy and PID were proposed as in [8] and [9] for robot arm control [10]. A detailed mathematical model of missiles and the analysis in using fuzzy logic as an autopilot for Proportional Navigation Guidance were presented [11]. The reference [12] proposed a guidance system using fuzzy logic. Path errors were used as the input of PD/PI controller and the outputs were combined and chosen using Fuzzy PI/PD-Weighting rule-base. Meanwhile, Fuzzy logic controller was used to replace the classical control [13]. A lateral-displacement controller using fuzzy were

represented for a roll-stabilized skid-to-turn missile. Sajad [14] showed a method using fuzzy gain schedule to control the lateral acceleration for 5DOF missile. The reference [15] used two-dimensional fuzzy PD and PI to control pitch angle to reduce the system overshoot. The combination controller was applied [16] but it only implement for a case of transfer function control roll channel using step command.

This paper focuses on designing and implementing Fuzzy-PI gyro-damping controller to simulate short-range air-to-surface missile using parameter data as in [5]. The current work is divided into five sections. General missile system architecture and classical controllers are described in section II. Section III introduce a proposed design and the detailed consideration controller as the rule-based, tuning mechanism. All results in simulation and comparison are shown in section IV. Finally, the conclusions are given in section V.

II. OVERVIEW OF HOMING MISSILE ARCHITETURE

A. Homing missile loop

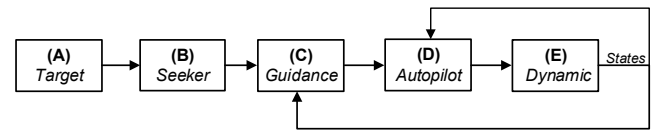


Fig. 1. The simulation process

The general architecture is mentioned as in [7]. The current work aims to design controller instead of studying in detail the structure and function of each block. Briefly, block A creates the target for interceptor missile. Block B track and transfer target's information to the missile continuously. A guidance block (block C) follows a guidance law to generate command signal for controller. This research uses common PN law in order to calculated command acceleration for missile guidance. The controller in Block D creates control signal used to deflect actuators of missiles. The detail of classical controller is presented in the next section. Block E describes equations of motion, numerically solve this derivative system to determine the states of object in every time step and then feedback as initial condition for next step.

To simplify the simulation and modeling, the missile is assumed to be a symmetrical rigid body via XZ body plane and all errors of actuators as well as sensors are ignored. The North-East-Down coordinate system is taken as the fixed frame in the earth. Euler angles are utilized to transform the reference frame to the body frame.

B. Classical controllers

Before designing the controller for any flying object, the non-linear aerodynamic coefficients are reduced the order and motion equations are linearized by using Laplace transformation. The desired acceleration command a_c following the PN law is calculated by the guidance block.

δ_e , q , a_m are elevator deflection, pitch rate and response acceleration respectively. The common PI/PID controller including three gains: Proportion (K_p), Integral (K_i) and Derivative (K_d) is described in Fig. 2.

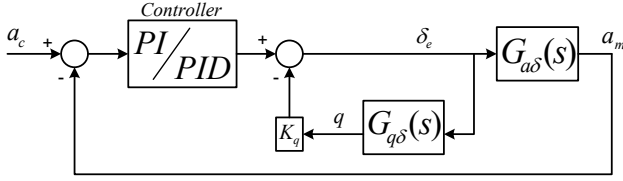


Fig. 2. PI/PID controller

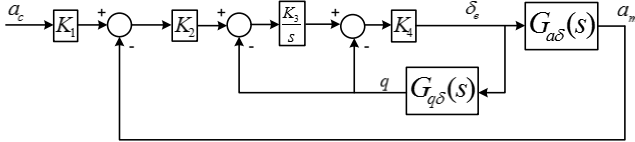


Fig. 3. Three-loop autopilot

The three-loop autopilot as shown in Fig. 3 consists of four gains, K_1 , K_2 , K_3 and K_4 . K_1 design depends on the rest of gains and the proportion between command and response acceleration. Due to the changing of transfer function of missile in every step, the parameters of controller are also varied respectively. Butterworth polynomial is used to find all these control gains combined with rate-gyro feedback for control homing missile.

III. FUZZY-PID GYRO-DAMPING CONTROLLER

Missile's control system is modeled for pitch and yaw channels to convert desired signal to actuators by guidance law. The fuzzy PI gyro-damping gain generate the control signal for the missile's elevators and rudders. The control's structure for pitch channel is shown in Fig. 4. In case of yaw channel, the structure is similar. The fuzzy logic is applied to adjust the parameters of PI controller in conjunction with gyro-damping gain based on analyzing the change of control gain to system's response.

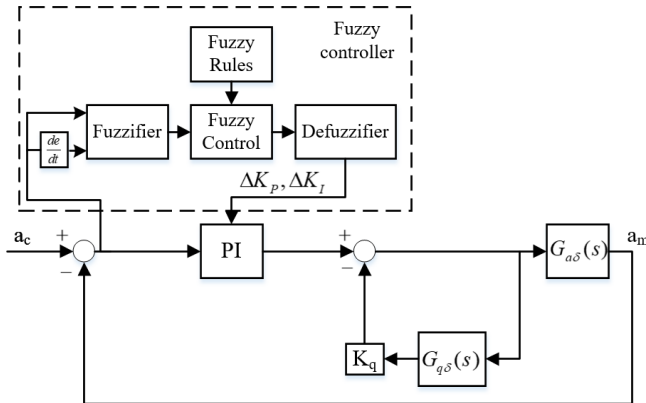


Fig. 4. The fuzzy PI gyro-damping controller

A. The PI combine with gyro-damping gain

The output equation of PI gyro-damping controller was described in equation (1).

$$\delta_e = K_p \cdot e + K_i \cdot \int e dt - K_q \cdot \dot{q} \quad (1)$$

As shown in figure 4, pitch rate feedback is added to inner loop which controls a short period damping and oscillation making the missile more stability.

The parameters K_p , K_i of PI controller are calculated as in equations (2) and (3)

$$K_p = K_{p0} + \Delta K_p \quad (2)$$

$$K_i = K_{i0} + \Delta K_i \quad (3)$$

In which: K_{p0} , K_{i0} , K_q are initial parameters calculated by using homogeneity method for denominator of transfer function and choosing polynomial.

B. Fuzzy logic

a) Fuzzifier

The input of fuzzy controller are errors and changing error of the system. The outputs are the amount of conversion control parameters. The inputs and outputs use seven fuzzy sets, corresponding to the linguistic variables: Negative Large (NL), Negative Medium (NM), Negative Small (NS), Medium (M), Positive Small (PS), Positive Medium (PM), Positive Large (PL). The domain of parameters is defined as $[a; b]$ and the linguistic variables are chosen as:

$$NL = [a; (5a+b)/6] \quad NS = [(5a+b)/6; (a+b)/6]$$

$$NM = [a; (3a+b)/6] \quad PM = [(a+3b)/6; b]$$

$$PL = [(a+5b)/6; b] \quad PS = [(a+b)/6; (a+5b)/6]$$

$$M = [(3a+b)/6; (a+3b)/6]$$

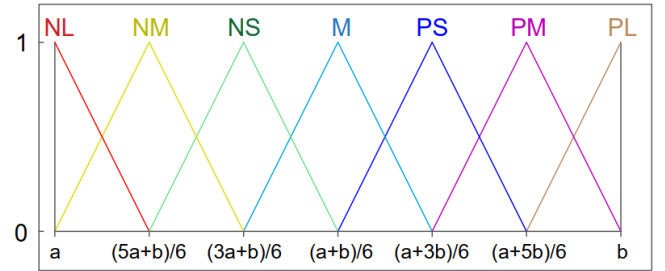


Fig. 5. Membership function

b) Fuzzy rules

The construction of the rule-base are based on relationship between control factor and system's response as shown in table I.

TABLE I. RELATIONSHIP PARAMETERS - SYSTEM RESPONSE

Gain	Change	Static Error	Time response	Overshoot
K_p	Increase	Decrease	Increase	Increase
K_i	Decrease	Vanish	Small Decrease	Increase

From the effect of parameters on the system as shown in Table I, the rule-base formation are described following principles: if the system's error is Negative Large, long time response {ER=NL, CER=NL}, K_p increases and K_i decreases by { ΔK_p =PL, ΔK_i =NL}. If overshoot is high {ER=PL,

CER=PL}, the amount of change are $\{\Delta K_p=AL, \Delta K_i=DN\}$.

According to this principle, the fuzzy rules consist of 49 law satisfying with this control system, as described in Table II. The first row and the first column refer to the linguistic variables of the ER and CER, the remaining cells show the linguistic variables of the K_p/K_i .

TABLE II. FUZZY RULES TABLE

ER \ CER	NL	NM	NS	M	PS	PM	PL
NL	PL/ NL	PL/ NL	PL/ NM	PL/ NM	PS/ NS	M/ M	M/ M
NS	PL/ NL	PL/ NL	PM/ NM	PS/ NS	PS/ NS	M/ M	NS/ NS
NM	PM/ NL	PM/ NM	PM/ NS	PS/ NS	M/ M	NS/ PS	NS/ PS
M	PM/ NM	PM/ NM	PS/ NS	M/ M	NS/ PS	NM/ PM	NM/ PM
PS	PS/ NM	PD/ NS	M/ M	NS/ PS	NS/ PS	NM/ PM	NM/ PL
PM	PS/ M	M/ M	NS/ PS	NM/ PS	NM/ PM	NM/ PL	NL/ PL
PL	M/ M	NS/ M	NS/ PM	NM/ PM	NM/ PM	NL/ PL	NL/ PL

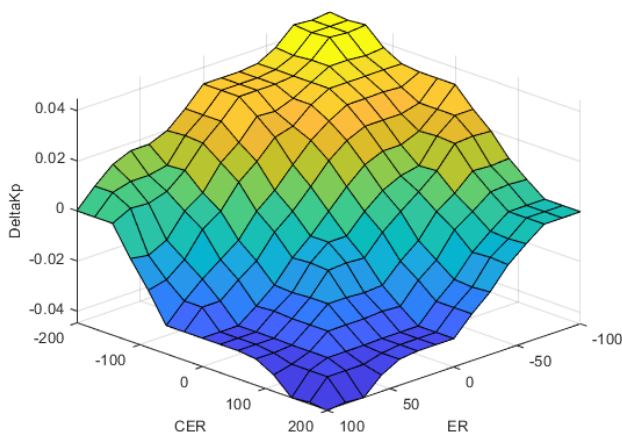


Fig. 6. Delta of K_p

Fig. 6 and Fig. 7 showed the output surface for each output ($\Delta K_p, \Delta K_i$) of the system versus two inputs (ER, CER). The fuzzy surface show an overview of the rule-base. The smoother surface is, the more accurate the rule-base system is.

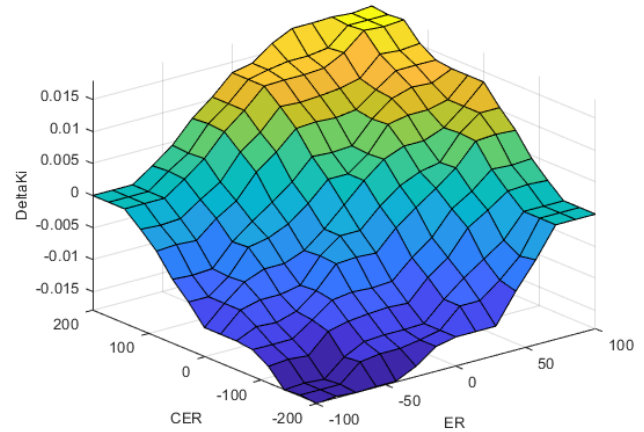


Fig. 7. Delta of K_i

c) Defuzzifier

A centroidal method is used to quantify results given by fuzzy sets and corresponding membership function. This is the most preferred method. It allows determine the quantifiable output with the participation of all fuzzy sets for output equally and accurately.

$$y' = \frac{\int_S y \mu_B(y) dy}{\int_S \mu_B(y) dy}$$

In which: S is defined as the domain of fuzzy volume B.

IV. RESULTS AND ANALYSIS

In this section, the behavior of hybrid controller is evaluated via performance and efficiency of missile. Characteristics of control model are mentioned as in [5]. Table III shows the initial conditions of missile and its target. The target is assumed to move on the ground with a velocity of 30 m/s. The missile is set to launch at an attitude of 1500m in Mach 1.

TABLE III. INITIAL CONDITIONS

	Missile	Target
Position (m)	[800 0 1500]	[2200 1500 0]
Velocity (m/s)	343	30
Angle (degree)	[0 5 2]	[0 0 10]

The requirement of system's responses is defined as: (1) overshoot under 10%, (2) the deflection of fins and acceleration in limited range beyond missile characteristics. Step response using combined Fuzzy and PI is compared with other controllers shown in Fig.8. Table 4 indicates clearly three-loop autopilot has highest rise time but lowest overshoot. Overshoot of PI controller is greater than 10%. The Fuzzy PI gyro-damping gain has good reaction for system. Between Fuzzy PI and three-loop, the overshoot is similar but the rise time and settling time of the former controller is better. Among three controllers, PI performs the worst response for the system with highest rise time and settling time. After testing characteristics of controllers with step command function, these controllers are applied to guided missile for both channel. Errors in acceleration are compared in Fig. 10 and Fig. 11.

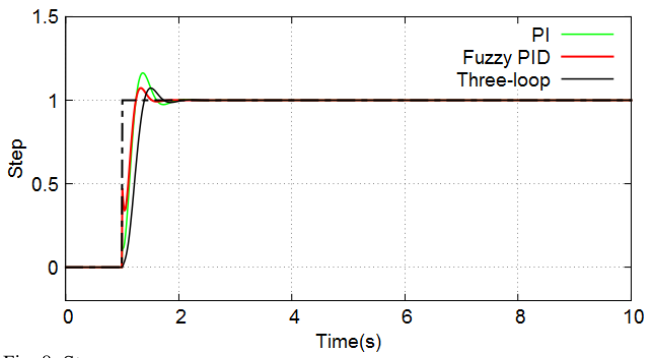


Fig. 8. Step responses

TABLE IV. COMPARISONS OF RESPONSES

	PI	Fuzzy PI	Three-loop
Overshoot (%)	16.30	7.3	7.2375
Rise Time (ms)	211.4	200	252.8
Settling Time (s)	0.8059	0.399	0.667

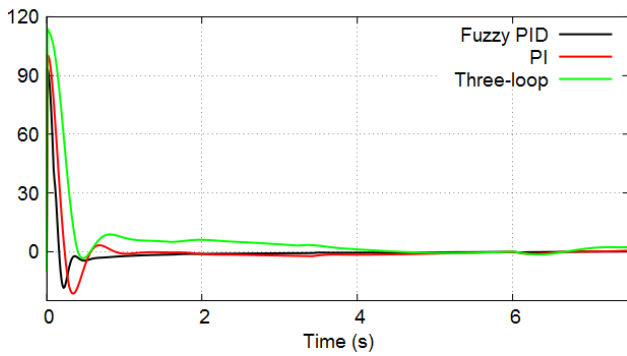


Fig. 9. Comparison error of controllers in pitch channel

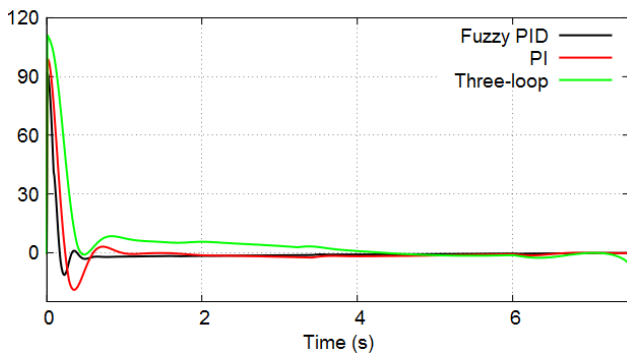


Fig. 10. Comparisons error of controllers in yaw channel

The detailed acceleration in pitch and yaw channel are shown in Fig. 11 and Fig.12. In this case, each channel acceleration is smaller than 30g ($g=9.81 \text{ m/s}^2$), which is limited range of the missile. The acceleration of missile meets the requirement. The range of rudders and elevators motion are $[-10^\circ \div 10^\circ]$ [5]. Deflection of fins described in Fig. 13 and Fig. 14, each response is in limited range. It demonstrated the effect of the new combined controller Fuzzy and PI with

Gyro-damping in this system. All responses are in safety zone and missile destroyed the target after nearly 7 seconds.

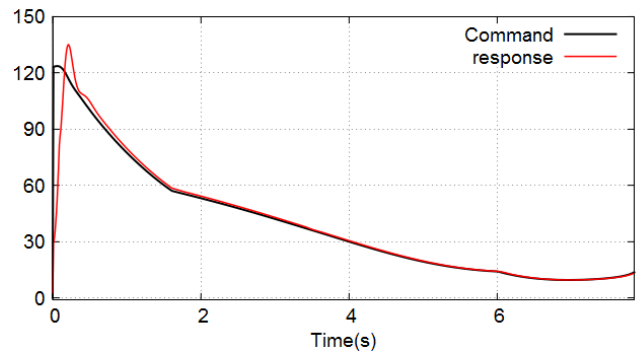


Fig. 11. Acceleration response in pitch channel

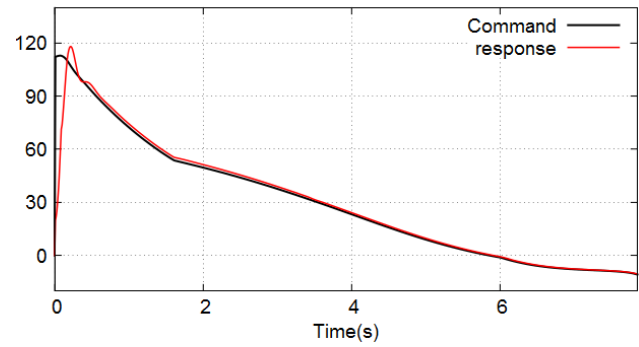


Fig. 12. Acceleration response in yaw channel

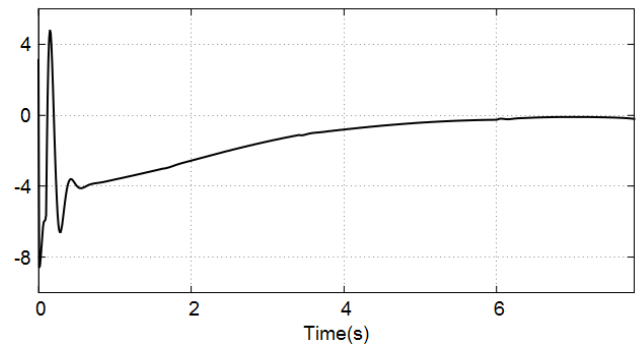


Fig. 13. Deflection of elevators

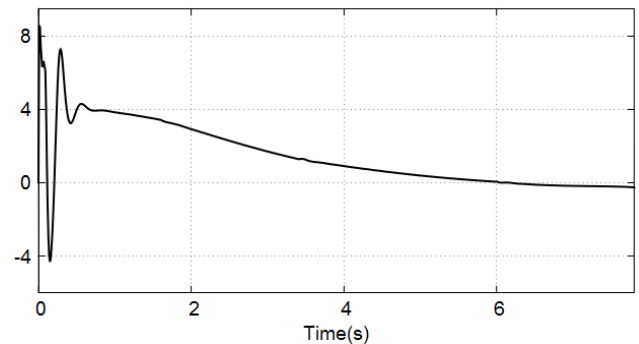


Fig. 14. Deflection of rudders

V. CONCLUSION

This research proposed a hybrid controller that combines the fuzzy logic and PI classical applying for guided missiles. The output of the controller depends on the initial coefficients from the traditional controller and the scaling factor defined by ER and the CER of process states. The HFPGD would ensure the missile could adapt, detect and track the target with full range of missiles operation. The results of comparison among three method in missile simulation show clearly that the proposed hybrid controller improved performance with small overshoot, especially in settling time and rise time.

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